

Generation Interconnection System Impact Study Report

For Queue Project Q13-0256

Mcconnico 138 kV

April 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0256. The Interconnection Customer proposes a 575.2 MW natural gas combined cycle facility interconnecting at the Mcconnico 138 kV substation in the Virginia area. The request was received on July 29, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 161 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$73,655,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The MCCONNICO – MANATI 8404 138.0 kV is overloaded to 265.0% of its 272 MVA emergency rating for the loss of the HUNTLEY – MCCONNICO 138.0 kV L. The project contributes 100.0% DFAX (575.2 MW) to this overload.

For the loss of MCCONNICO – HUNTLEY 138.0 kV L, the MANATI 8404 – DOUBLE TOLL GATE 138.0 kV loads to 251.8% of its 282 MVA rating. Project DFAX: 100.0%.

The HUNTLEY – TWIN LAKES ALTE 138.0 kV is overloaded to 184.5% of its 387 MVA emergency rating for the loss of the MANATI 8404 – MCCONNICO 138.0 kV. The project contributes 100.0% DFAX (575.2 MW) to this overload.

The HUNTLEY – MCCONNICO 138.0 kV is overloaded to 182.8% of its 394 MVA emergency rating for the loss of the MANATI 8404 – MCCONNICO 138.0 kV. The project contributes 100.0% DFAX (575.2 MW) to this overload.

The DOUBLE TOLL GATE – SPRINGFIELD 138.0 kV is overloaded to 159.6% of its 256 MVA emergency rating for the loss of the MCCONNICO – HUNTLEY 138.0 kV L. The project contributes 56.3% DFAX (323.8 MW) to this overload.

The NEW ROAD – AG RELIANT 138.0 kV L is overloaded to 138.1% of its 277 MVA emergency rating for the loss of the LOUDON 345 kV/138 kV Auto. The project contributes 9.2% DFAX (53.0 MW) to this overload.

The HUNTINGDON DISTRICT – AG RELIANT 138.0 kV L is overloaded to 131.1% of its 266 MVA emergency rating for the loss of the LOUDON 345/138 kV TX. The project contributes 9.2% DFAX (53.0 MW) to this overload.

The DOUBLE TOLL GATE – ABBOTSFORD 138.0 kV is overloaded to 127.8% of its 221 MVA emergency rating for the loss of the MCCONNICO – HUNTLEY 138.0 kV L. The project contributes 43.7% DFAX (251.4 MW) to this overload.

ARCOLA – TWIN LAKES ALTE 138.0 kV: 123.9% loading (338 MVA) following the outage of the LOUDON 345/138 kV XFMR; DFAX = 9.2%.

The NEW ROAD – ARCLA 138.0 kV is overloaded to 121.1% of its 325 MVA emergency rating for the loss of the LOUDON 345-138 kV Transformer Bank. The project contributes 9.2% DFAX (53.0 MW) to this overload.

For the loss of ARCOLA – TWIN LAKES ALTE 138.0 kV, the WHEELER – MAMMOTH PACIFIC 2 138.0 kV loads to 113.2% of its 268 MVA rating. Project DFAX: 34.8%.

The MAMMOTH PACIFIC 2 – T.S. POWER 138.0 kV is overloaded to 110.7% of its 297 MVA emergency rating for the loss of the TWIN LAKES ALTE – ARCOLA 138.0 kV. The project contributes 34.8% DFAX (200.1 MW) to this

overload.

The T.S. POWER – TWIN LAKES ALTE 138.0 kV L is overloaded to 109.1% of its 320 MVA emergency rating for the loss of the ARCOLA – TWIN LAKES ALTE 138.0 kV. The project contributes 34.8% DFAX (200.1 MW) to this overload.

For the loss of KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345.0 kV, the YAMHILL – 3 MILE KNOLL 138.0 kV loads to 108.9% of its 267 MVA rating. Project DFAX: 5.3%.

The COW – YAMHILL 138.0 kV is overloaded to 106.9% of its 267 MVA emergency rating for the loss of the INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345.0 kV L. The project contributes 5.3% DFAX (30.6 MW) to this overload.

For the loss of KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345.0 kV, the COW – FEAGAN'S MILL 138.0 kV line loads to 105.6% of its 260 MVA rating. Project DFAX: 5.3%.

CASCADE STEEL – FEAGAN'S MILL 138.0 kV: 105.6% loading (255 MVA) following the outage of the INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345.0 kV; DFAX = 5.3%.

NOVEC PARADISE – DEAN DRIVE AREA ELECTRIC GENERATORS 138.0 kV Line: 104.5% loading (204 MVA) following the outage of the BLAINE – CHAFFEE GAS RECOVERY 138.0 kV; DFAX = 6.3%.

The W STAUNTON – DEAN DRIVE AREA ELECTRIC GENERATORS 138.0 kV line is overloaded to 104.4% of its 204 MVA emergency rating for the loss of the BLAINE – CHAFFEE GAS RECOVERY SUBSTATION 138.0 kV. The project contributes 6.3% DFAX (36.3 MW) to this overload.

KITIMAT SUBSTATION – CASCADE STEEL 138.0 kV: 103.9% loading (247 MVA) following the outage of the KEARNEYVILLE – INTERNATIONAL PAPER VALLIANT OK 345.0 kV L; DFAX = 5.3%.

The JET – HUNTINGDON DISTRICT 138.0 kV is overloaded to 103.4% of its 195 MVA emergency rating for the loss of the HUNTINGDON DISTRICT – CLEBURNE 138.0 kV Line. The project contributes 14.9% DFAX (85.7 MW) to this overload.

For the loss of INTERNATIONAL PAPER VALLIANT OK – KEARNEYVILLE 345.0 kV line, the TERROR LAKE MICROGRID – KITIMAT 138.0 kV loads to 101.1% of its 244 MVA rating. Project DFAX: 5.3%.

HUNTINGDON DISTRICT – CLEBURNE 138.0 kV: 100.0% loading (202 MVA) following the outage of the HUNTINGDON DISTRICT – JET 138.0 kV L; DFAX = 14.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MANATI 8404 – MCCONNICO 138.0 kV	Rebuild line with higher-capacity conductor	\$5,546,000	79% (\$4,390,000)
MANATI 8404 – DOUBLE TOLL GATE 138.0 kV	Rebuild line with higher-capacity conductor	\$9,181,000	79% (\$7,268,000)
HUNTLEY – TWIN LAKES ALTE 138.0 kV	Rebuild line with higher-capacity conductor	\$32,270,000	79% (\$25,546,000)
MCCONNICO – HUNTLEY 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,917,000	79% (\$4,684,000)
DOUBLE TOLL GATE – SPRINGFIELD 138.0 kV	Rebuild line with higher-capacity conductor	\$1,896,000	79% (\$1,501,000)
NEW ROAD – AG RELIANT 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,539,000	50% (\$1,753,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUNTINGDON DISTRICT – AG RELIANT 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,310,000	50% (\$2,630,000)
DOUBLE TOLL GATE – ABBOTSFORD 138.0 kV	Rebuild line with higher-capacity conductor	\$7,788,000	79% (\$6,165,000)
ARCOLA – TWIN LAKES ALTE 138.0 kV	Rebuild line with higher-capacity conductor	\$3,524,000	50% (\$1,745,000)
NEW ROAD – ARCLA 138.0 kV	Rebuild line with higher-capacity conductor	\$2,734,000	50% (\$1,354,000)
WHEELER SUBSTATION – MAMMOTH PACIFIC 2 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,245,000	66% (\$4,100,000)
MAMMOTH PACIFIC 2 – T.S. POWER 138.0 kV	Rebuild line with higher-capacity conductor	\$4,089,000	66% (\$2,685,000)
T.S. POWER – TWIN LAKES ALTE 138.0 kV Line	Reconductor line	\$1,357,000	66% (\$891,000)
YAMHILL – 3 MILE KNOLL 138.0 kV	Reconductor line	\$1,873,000	65% (\$1,224,000)
YAMHILL – COW 138.0 kV	Reconductor line	\$676,000	65% (\$442,000)
COW – FEAGAN'S MILL 138.0 kV	Reconductor line	\$1,594,000	65% (\$1,041,000)
CASCADE STEEL – FEAGAN'S MILL SUBSTATION 138.0 kV	Reconductor line	\$976,000	65% (\$638,000)
NOVEC PARADISE – DEAN DRIVE AREA ELECTRIC GENERTORS 138.0 kV Line	Reconductor line	\$1,441,000	63% (\$913,000)
W STAUNTON – DEAN DRIVE AREA ELECTRIC GENERATORS 138.0 kV	Reconductor line	\$351,000	63% (\$223,000)
KITIMAT – CASCADE STEEL 138.0 kV	Reconductor line	\$607,000	65% (\$396,000)
JET – HUNTINGDON DISTRICT 138.0 kV	Reconductor line	\$4,018,000	70% (\$2,801,000)
KITIMAT – TERROR LAKE MICROGRID 138.0 kV	Reconductor line	\$1,329,000	65% (\$868,000)
HUNTINGDON DISTRICT – CLEBURNE 138.0 kV	Reconductor line	\$575,000	69% (\$397,000)
	Total allocated to project		\$73,655,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.